National College of Ireland

Project Submission Sheet

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Lecturer:	Christian Horn		
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Signature:	Ibraheem
Date:	23/03/2026

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4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
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Your Name/Student Number	Course	Date
Hafiz Muhammad Ibraheem	Statistics	23/03/2026

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AI Acknowledgment

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Tool Name	Brief Description	Link to tool
Claude	Used for the idea and to understand the assignment.	

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used**.

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Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[Place evidence here]

Additional Evidence:

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Predicting Fish Weight Using Regression Models: A Comparative Study on the Fish Market Dataset

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***Abstract*—This paper analyzed Fishmarket Database fish, including 158 separate observations from 7 species of Fresh Water Fish caught in Northern Europe (weight, total length, height and width measurements). An erroneous observation (weight of 0 grams) was excluded before finalizing the dataset to provide 157 usable observations because 2 of the 3 length measurements (correlation coefficient larger than 0.99) were highly correlated with each other. Three regression models were created to predict the weight of fish based on the total length (Length3), height, width and species: standard multiple linear regression (RMSE of 91.81 grams), linear regression with pairwise interaction terms (RMSE of 51.74 grams), and logarithmic regression (RMSE of 50.70 grams). The best model for predicting the weight of fish was the logarithmic regression because it produced the lowest RMSE and had a lower degree of variation in its residuals. A Perch with a height of 12.8 cm, width of 6.9 cm and total length of 41.9 cm would weigh approximately 907.62 grams.**

***Index Terms*—regression analysis, fish weight prediction, linear regression, interaction model, log-log regression, RMSE, multicollinearity**

I. Introduction

Accurate weight estimation of fish from morphological measurements is critical in fisheries management and aquaculture, as well as in the ecological studies of the species studied. This research will develop and compare regression models for predicting the weight of freshwater fish from physical (morphological) measurements and by the species.

The dataset used in this analysis consists of 159 records; the records contained in the dataset are found in the file fishmarket.csv. The dataset has the following attributes: Species (categoric), weight (g), Standard Length (Length1), Fork Length (Length2), Total Length (Length3), width, and height (in cm). The variable of interest for each fish is the weight.

The process used to build the regression models and evaluate their validity includes:

1) Carrying out exploratory data analysis (EDA) using descriptive statistics (i.e., summary data) data visualization (e.g., graphs, charts, etc.), data cleaning, and assessing for multicollinearity
2) Building and evaluating three different types of regression models, (a) linear regression, (b) linear regression with an interaction term, and (c) log-log regression
3) Evaluating the models and determining the final weight estimate using Root Mean Square Error (RMSE) values and the final weight estimate for a specific perch.

II. Exploratory Data Analysis

A. Dataset Overview

The dataset that is used in this analysis contains 159 rows, 7 columns, no missing values, and no duplicate records. Seven species are present:

1) Perch (56)
2) Bream (35),
3) Roach (19)
4) Pike (17)
5) Smelt (14)
6) Parkki (11)
7) and Whitefish (6).

Table I summarises the descriptive statistics for all numeric variables.

TABLE I: Descriptive Statistics ($n = 159$)

Variable	Mean	Std	Min	75%	Max
Weight (g)	398.3	358.0	0.0	650.0	1650.0
Length1 (cm)	26.2	10.0	7.5	32.2	59.0
Length2 (cm)	28.4	10.7	8.4	34.7	63.4
Length3 (cm)	31.2	11.6	8.8	37.8	68.0
Height (cm)	8.97	4.29	1.73	12.37	18.96
Width (cm)	4.42	1.69	1.05	5.58	8.14

B. Removal of Erroneous Records

Inspection of the `Weight` variable revealed one record with Weight $= 0$ g, which is physically impossible. This observation was removed (notebook Cell 12), leaving **158 valid records** for all subsequent analysis.

C. Multicollinearity Among Length Variables

All three measurements for length were very highly correlated (all pairs of correlation coefficients are ¿ 0.99), as evidenced by both the scatter plot matrix (shown in Fig. 1) and the correlation heatmap (Table II). Using all three variables in a regression analysis would create considerable multicollinearity, which would inflate standard errors and render coefficient estimates unstable. Therefore, only Length3 (total length; r = 0.923 with weight) was maintained in the analysis; Length1 and Length2 were removed (cell 17).

TABLE II: Correlation Matrix — Length Variables

	Length1	**Length2**	**Length3**
Length1	1.0000	0.9995	0.9920
Length2	0.9995	1.0000	0.9941
Length3	0.9920	0.9941	1.0000

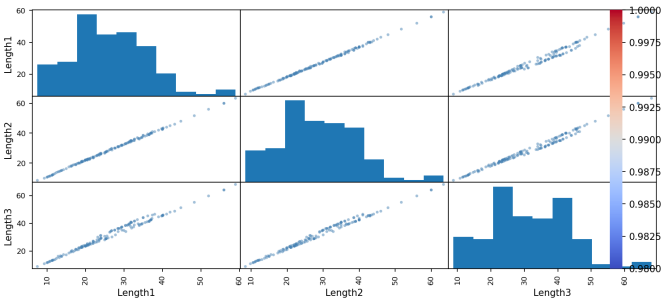

Fig. 1: Scatter matrix of Length1, Length2, Length3 — all pairs $r > 0.99$, confirming extreme multicollinearity.

D. Correlations with Weight

After reducing to the final predictor set, correlations with Weight are: Length3 ($r = 0.923$), Width ($r = 0.887$), and Height ($r = 0.724$). The heatmap (Fig. 2) Verify all three predictors share strong positive correlation with Weight.

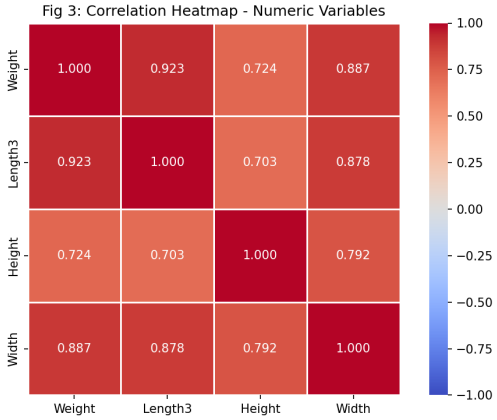

Fig. 2: Correlation heatmap of the four numeric variables retained after multicollinearity reduction.

E. Distribution of Weight and Scatter Plots

The distribution of raw weight is very non-normal and skewed to the right, as shown by the histogram on the left in Fig. 3. After transformation by the natural logarithm, the distribution appears to be approximately symmetric, providing justification for the log-log regression model. The scatterplots in Fig. 4 show that the relationship between weight and each predictor is curved and non-linear, thus further supporting the use of logarithmic transformation.

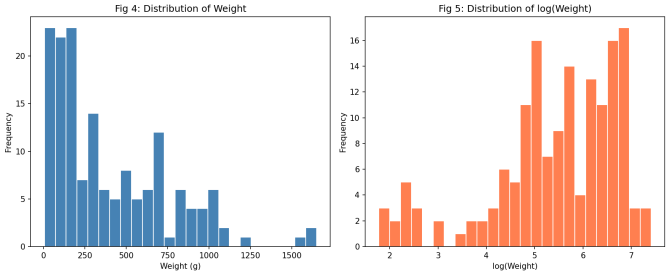

Fig. 3: Distribution of Weight (left, right-skewed) and log(Weight) (right, approximately symmetric).

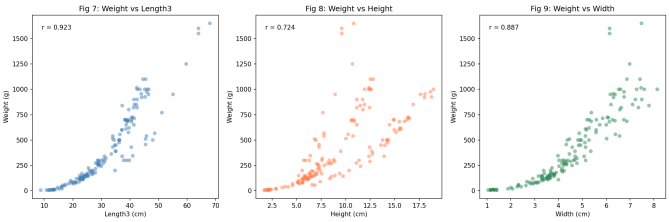

Fig. 4: Scatter plots of Weight vs Length3 ($r = 0.923$), Height ($r = 0.724$), and Width ($r = 0.887$).

III. Regression Models

A. Model 1: Multiple Linear Regression

The first model regresses `Weight` on `Length3`, `Height`, `Width`, and `Species` (dummy-coded, Bream as reference):

$$\hat{W} = \beta_0 + \beta_1 L_3 + \beta_2 H + \beta_3 W_d + \sum_s \beta_s \cdot \text{Species}_s + \varepsilon \quad (1)$$

The model produced an R-squared value of 0.934 (adjusted R-squared of 0.930) and a significantly high F-statistic value (F=231.6, p¡0.001). The variable length3 showed a very strong statistical significance (coefficient of 36.14, p¡0.001) while height and width did not show significant results at a 5% confidence level (p=0.490 for height and p=0.845 for width). Of the dummy variable species classification, smelt and parkki were shown to differ the most from that of bream, while pike, roach, and whitefish were not found to be statistically significant. The model's root mean square error was 91.81 g. The residuals versus fitted values plot exhibit some clear violations of ordinary least squares (OLS) assumptions; there is a clear fan shape indicating heteroscedasticity in the residuals vs. fitted values plot and very heavy tails in the Q-Q plot (skewness of 1.11 and kurtosis of 5.84). The evidence of the violation of OLS assumptions supports the need for the development of more flexible model specifications.

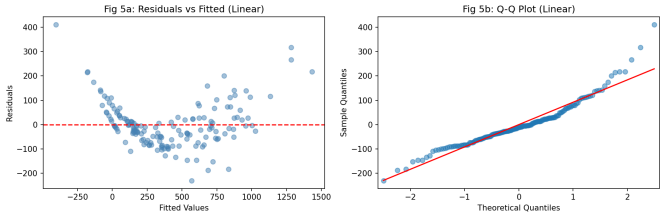

Fig. 5: Model 1 diagnostics: Residuals vs. Fitted (left) and Normal Q-Q plot (right). Clear heteroscedasticity and heavy tails are visible.

B. Model 2: Linear Regression with Interactions

Pairwise interaction terms capture non-additive effects between continuous predictors — a longer and wider fish grows disproportionately heavier:

$$\begin{aligned}\hat{W} = \beta_0 + \beta_1 L_3 + \beta_2 H + \beta_3 W_d \\ + \beta_4(L_3 \cdot H) + \beta_5(L_3 \cdot W_d) + \beta_6(H \cdot W_d) \\ + \sum_s \beta_s \cdot \text{Species}_s + \varepsilon \end{aligned} \tag{2}$$

R squared = 0.979, Adj. R squared = 0.977, F-statistic = 561.7, p < 0.001. There was significant interaction between L3 and Wd (coefficient = 4.91, p < 0.001) suggesting that both length and width have a joint influence. The other interactions and most of the main effects are not significant individually. A high condition number of 1.4810^4 indicates numerical multicollinearity as a result of the inclusion of interaction terms. RMSE = 51.74 grams, representing a 43.7% improvement over Model One. The diagnostic plots provided (Fig. 6) illustrate an improvement but still do not represent ideal conditions. The fan pattern has reduced; however there are still heavy tails in the Q-Q plot (Kurtosis 9.23).

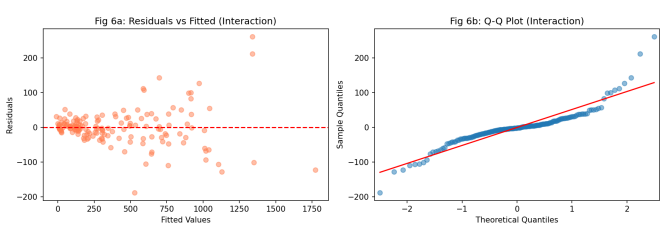

Fig. 6: Model 2 diagnostics: Residuals vs. Fitted (left) and Normal Q-Q plot (right). Reduced heteroscedasticity but heavy tails persist.

C. Model 3: Logarithmic (Log-Log) Regression

The relationship between the dimensions of fish, as expressed by the volume proportionate to their length, follows a power law. When you convert this relationship to logarithmic terms, you find a linear regression that will provide these weights. A visual representation of the normal distribution can be found in Figure 3.

This is mathematically demonstrated by the following summary statistics:

$$\log(\hat{W}) = \beta_0 + \beta_1 \log(L_3) + \beta_2 \log(H) + \beta_3 \log(W_d) + \sum_s \beta_s \cdot \text{Species}_s + \varepsilon \tag{3}$$

The R-squared is 0.996 (Adjusted R-squared = 0.996) and F = 3939 and p < 0.001 for all three predictors that are in log form; therefore, each are defined at the 0.0001 significance level. When adding up the three exponents involved (i.e., 1.794 + 0.706 + 0.526 = 3.03), this value will be very close to an actual constant of 3 and would produce cubic volume scaling characteristics. The condition number was 208, indicating that this calculation was not influenced by numerical instability. The complete coefficients can be found in Table III.

TABLE III: Log-Log Model Coefficients ($R^2 = 0.996$, RMSE $= 50.70$ g)

Parameter	Coef.	Std Err	t	p-value
Intercept	-2.967	0.268	-11.06	< 0.001
C(Sp)[Parkki]	0.156	0.035	4.44	< 0.001
C(Sp)[Perch]	0.260	0.069	3.76	< 0.001
C(Sp)[Pike]	0.129	0.129	1.00	0.319
C(Sp)[Roach]	0.153	0.064	2.40	0.018
C(Sp)[Smelt]	0.033	0.110	0.30	0.763
C(Sp)[Whitefish]	0.279	0.065	4.30	< 0.001
log(Length3)	1.794	0.155	11.58	< 0.001
log(Height)	0.706	0.149	4.75	< 0.001
log(Width)	0.526	0.114	4.64	< 0.001

RMSE $= 50.70\,\mathbf{g}$ — the best of all three models. The diagnostic plots (Fig. 7) show the most regular residual behaviour: approximately homoscedastic residuals and the Q-Q plot closely following the normal line.

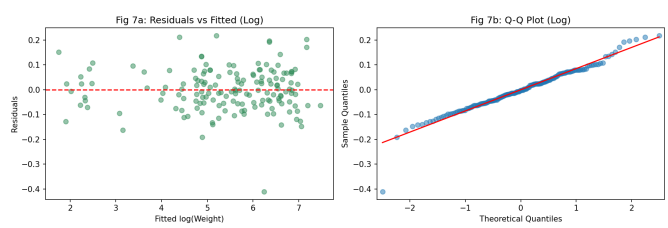

Fig. 7: Model 3 diagnostics: Residuals vs. Fitted log(Weight) (left) and Normal Q-Q plot (right). Residuals are approximately homoscedastic and near-normal.

IV. Conclusion

A. Model Comparison

The three models are summarised in Table IV and illustrated in Fig. 8. Of these, the logarithmic model has the best performance with respect to RMSE (50.70 g), Adj. R2 (0.996), well condition design matrix and reliable residual diagnostic results compared to the other two models. The interaction model performed slightly worse than the logarithmic model for RMSE (51.74 g); however, it had issues with multi-collinearity (condition number 1.48 × 104) and more heavy-tailed residuals.

The linear model is by far the weakest, exhibiting a high RMSE (91.81 g) due to the presence of heteroscedasticity.

TABLE IV: Model Comparison Summary

Model	**Adj.** R^2	**RMSE (g)**	**Cond. No.**
Linear	0.930	91.81	936
Interaction	0.977	51.74	14,800
Log-Log	0.996	50.70	208

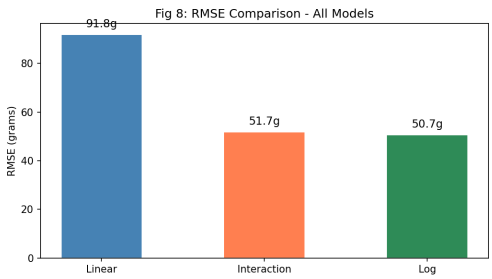

Fig. 8: RMSE comparison across all three regression models.

B. Best Model Selection

Logarithmic (log-log) regression model was chosen as most appropriate since:

1) Its RMSE is 50.70 g - the lowest of all models tested.
2) It has the highest explanatory power with an adjacent R2 = 0.996.
3) Weight/mass is proportional to the cubed value of the linear dimensions of that object (Sum of the exponentials is approximately 3.03).
4) All predictor variables used within this model are has shown to be individually significant using the p-value cut-off point of p ¡ 0.05.
5) Well-conditioned designed regression heuristics using the condition number (C = 208).
6) The logarithmic model has provided the best residual diagnostics. The models exhibited nearly homoscedasticity as well as approximated normality.

C. Predicted Weight of the Test Perch

By using the logarithmic model (Eq. 3), the weight of a Perch with Height $= 12.8\,\text{cm}$, Width $= 6.9\,\text{cm}$, and Length3 $= 41.9\,\text{cm}$ was computed as follows:

$$\begin{aligned} \log(\hat{W}) &= -2.967 + 1.794\log(41.9) \\ &\quad + 0.706\log(12.8) + 0.526\log(6.9) + 0.260 \\ \hat{W} &= e^{\log(\hat{W})} \end{aligned} \tag{4}$$

$$\boxed{\hat{W} \approx 907.62 \text{ grams}} \tag{5}$$

This prediction is consistent with the observed weight range for Perch in the dataset ($\approx 5.9\,\text{g}$ to $1000\,\text{g}$) and is plausible for the stated dimensions.